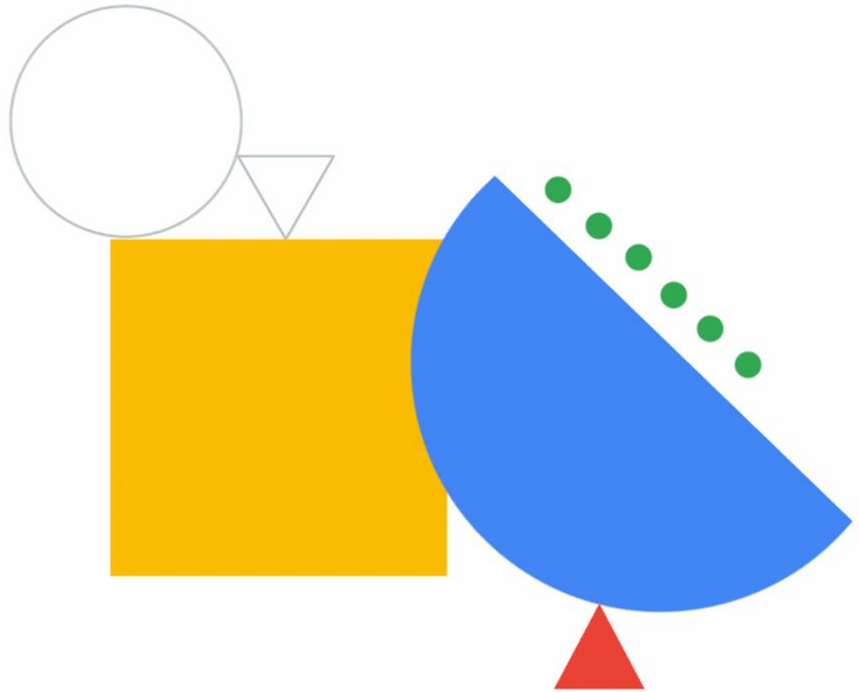
Google Cloud

Alerting Policies



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Objectives

01 Explain why SLI, SLO, and SLA are important.

02 Explain alerting policies and alerting strategies.

03 Explain error budgets.

04 Identify types of alerts and common uses for each.

05 Use Cloud Monitoring to manage services.



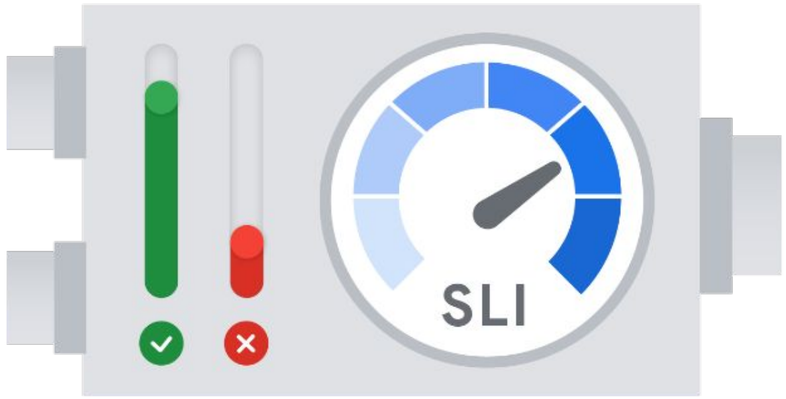
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In this section, you explore



- SLI, SLO, and SLA
- Developing an alerting strategy
- Creating alerts
- Service Monitoring

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Service Level Indicator

Carefully selected monitoring metrics that measure one aspect of a service's reliability

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Service Level Objective

Combines a service level indicator with a target reliability and will generally be somewhere just short of 100%, for example, 99.9% ("three nines")

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Service Level Objective

- **S** Specific
- **M** Measurable
- **A** Achievable
- **R** Relevant
- **T** Time-bound



Service Level Agreement

Commitments made to your customers that your systems and applications will have only a certain amount of down time

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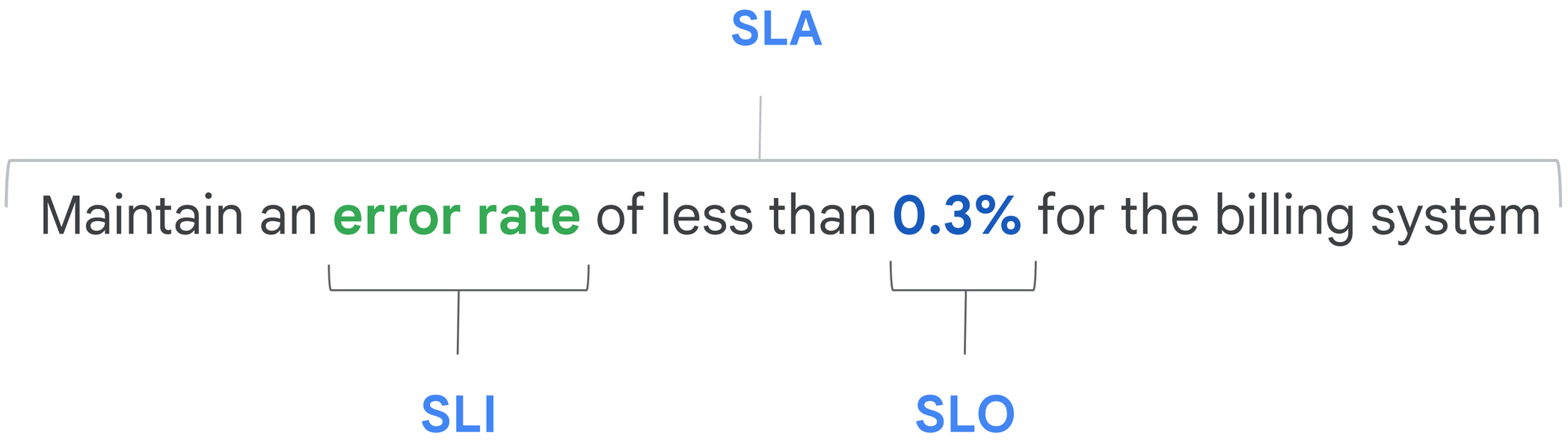




To improve service reliability, all parts of the business must agree that these are an accurate measure of user experience and must agree to use them as a primary driver for decision making

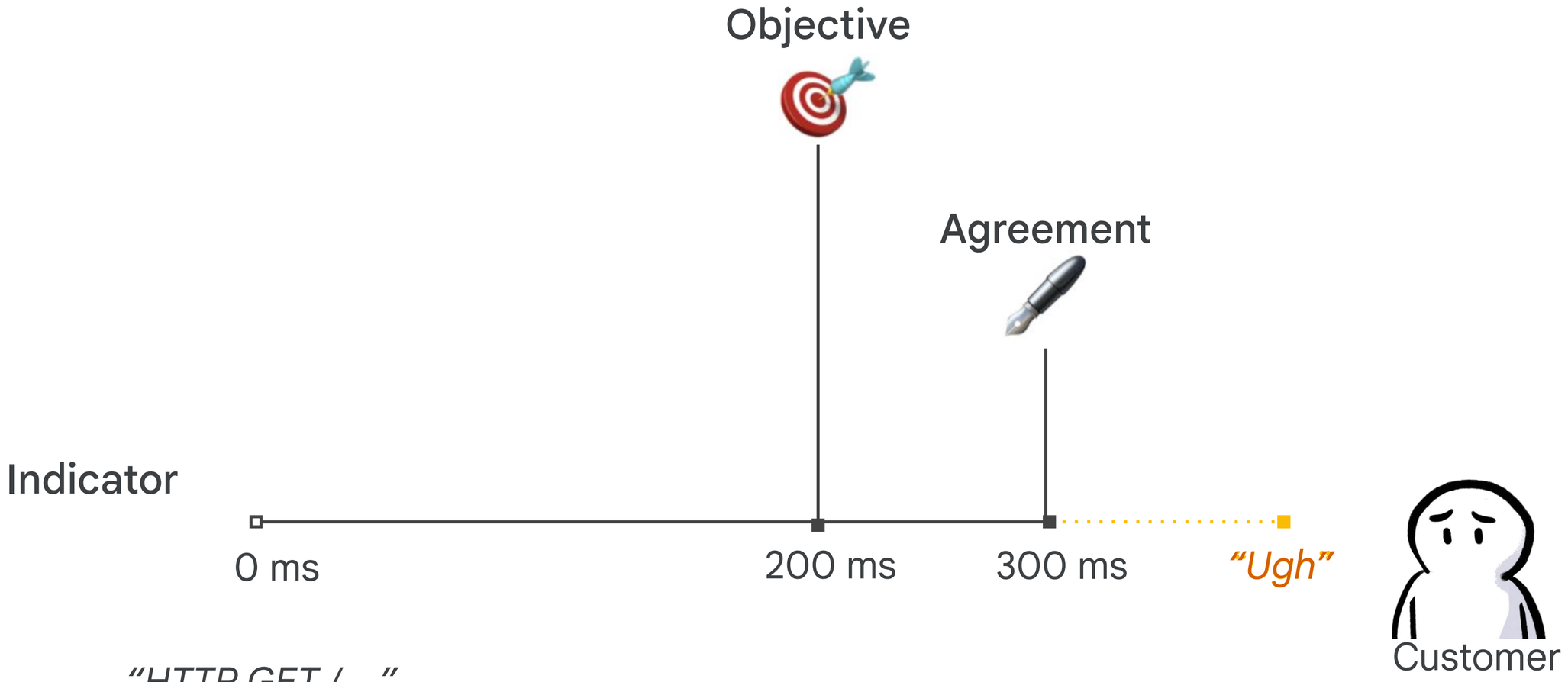
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SLI, SLO and SLA example



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Deciding the SLA, SLO and SLI



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In this section, you explore

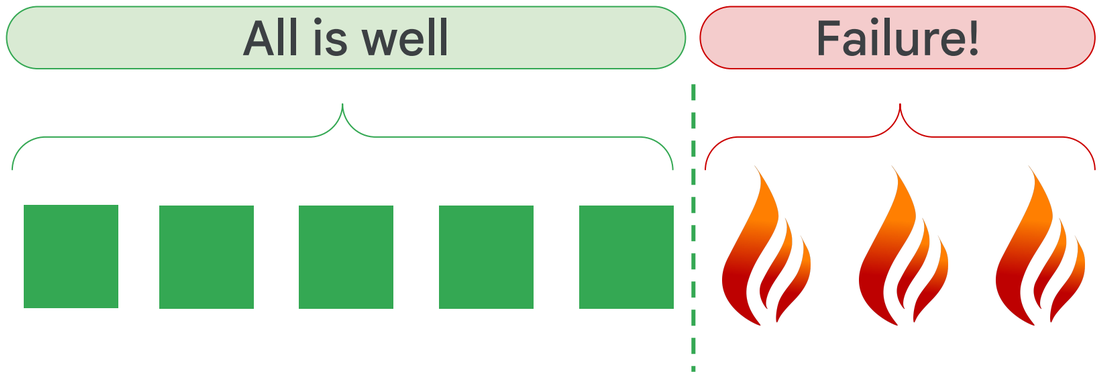


- SLI, SLO, and SLA
- **Developing an alerting strategy**
- Creating alerts
- Service Monitoring

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Goal: Person is notified when needed

- A service is down.
- SLOs or SLAs are heading toward not being met.
- Something needs to change.



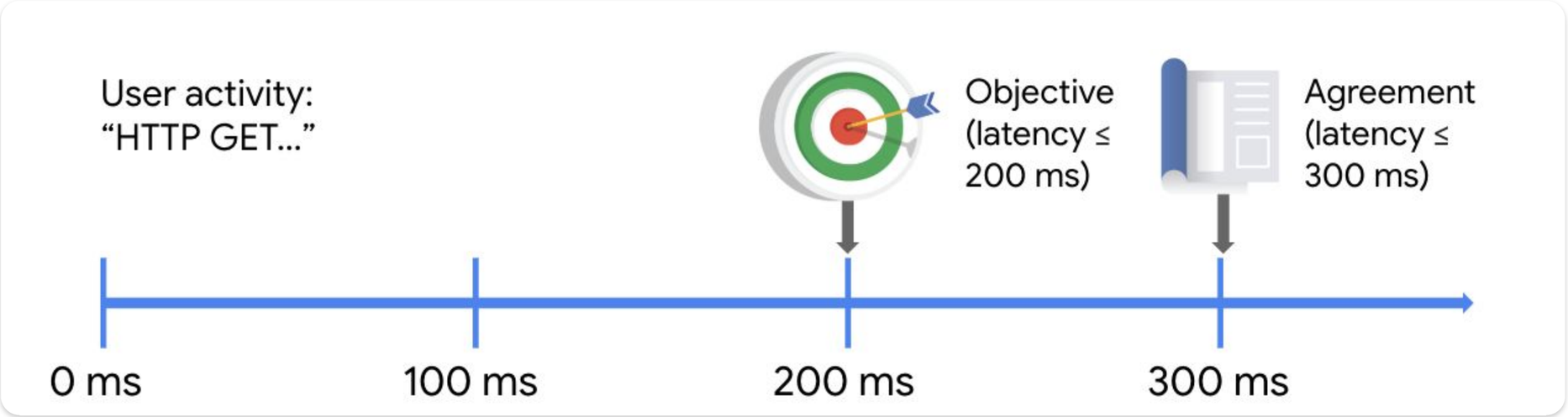
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When something puts the error budget in danger: Alert!

Error budget = Perfection - SLO

- **SLIs** are the things you measure.
- **SLOs** represent an achievable target.

If the SLO is: “90% of requests must return in 200 ms,” then the error budget is: 100% - 90% = 10%



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Evaluating alerts

Precision	Recall	Detection time	Reset time
Relevant alerts = Relevant alerts + irrelevant alerts	Relevant alerts = Relevant alerts + missed alerts	How long it takes the system to notice an alert condition	How long alerts fire after an issue is resolved
Adversely affected by false positives	Striving for precision might cause events to be missed	Long detection times can negatively affect the error budget	Continued alerts on repaired systems can lead to confusion
	Recall is adversely affected by missed alerts	Raising alerts too fast may result in poor precision	

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When error count is trending greater than error budget: Alert!

- Windows, as a concept related to SRE and alerting, is frequently used in two main ways:
  - An SLO of 99.9% is meaningless without a time period.
  - A **Window** is the period that the error calculation is made over.

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Alert window lengths

Small windows

- Faster alert detection
- Shorter reset time
- Poor precision

For a 99.9% availability SLO over 30 days, a 10-minute window would alert in 0.6 seconds if a full outage occurs.

Consume only 0.02% of the error budget.

Longer windows

- Better precision
- Longer reset and detection times
- More error budget spent before alert

For a 99.9% availability SLO over 30 days, a 36-hour window would alert in 2 minutes 10 seconds if a full outage occurs.

Represent 5% of the error budget.

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Try short windows with successive failure counts

An error is spotted quickly but treated as an anomaly until **three** windows fail in a row.

Recall becomes worse:

- If the duration is 10 minutes, a 100% outage for 5 minutes is not detected.
- If errors spike up and down, they might never be detected.

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Use multiple conditions for better precision and recall

- Many variables can affect a good alerting strategy:
  - Amount of traffic
  - Error budget
  - Peak and slow periods
- You can define multiple conditions in an alerting policy to try to get better precision, recall, detection time, and rest time.
- You can also define multiple alerts through multiple channels:
  - Automated and human
- See the SRE Workbook for more information: https://landing.google.com/sre/workbook/chapters/alerting-on-slos/

Prioritize alerts based on customer impact and SLA

- Involve humans only for critical alerts.
- Use severity level to assess the priority of an alert.
- Configure how to triage low and high priority alerts.



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In this section, you explore



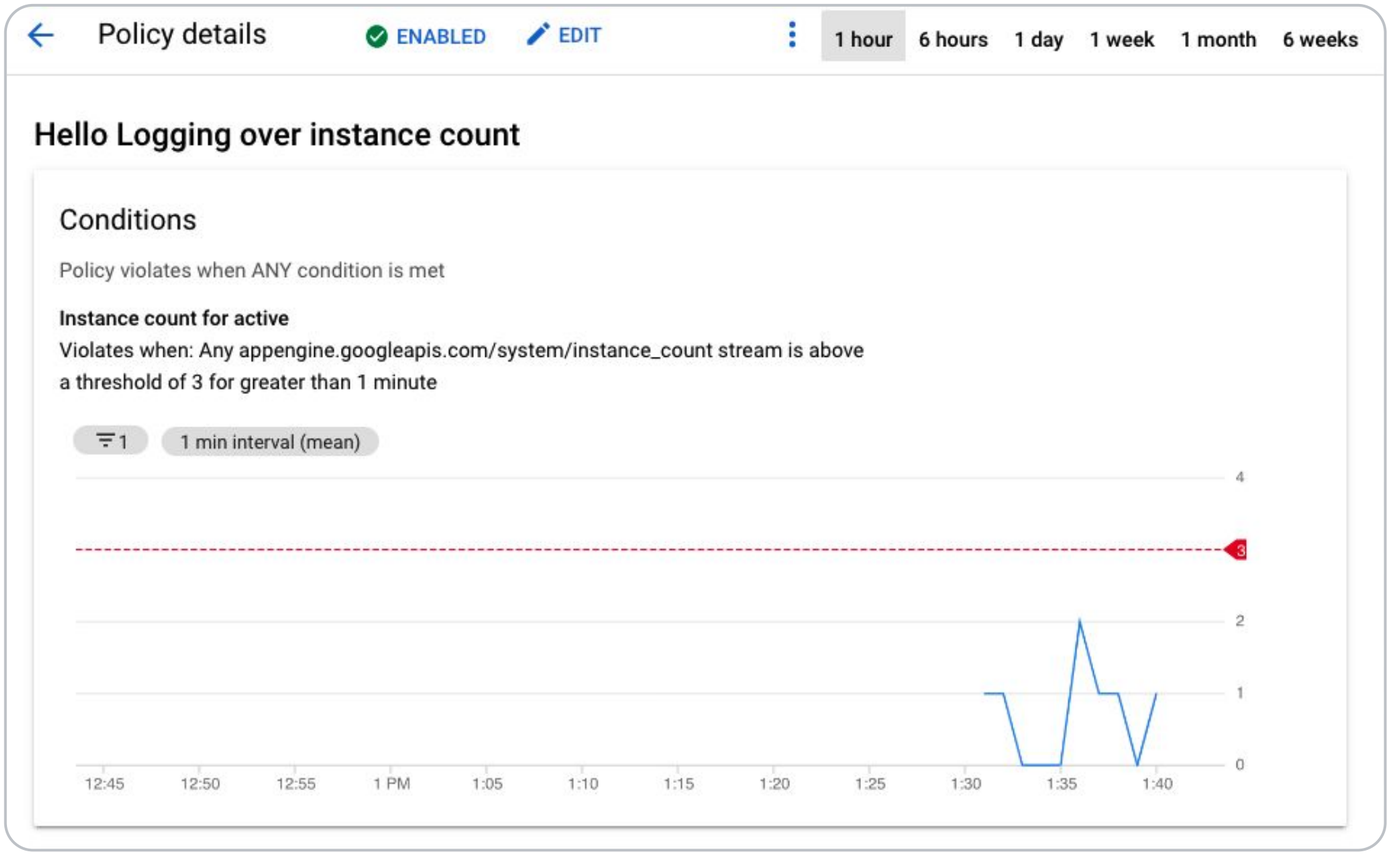
- SLI, SLO, and SLA
- Developing an alerting strategy
- **Creating alerts**
- Service Monitoring

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Use alerting policies to define alerts

An alerting policy has:

- A name
- One or more conditions
- Notifications
- Documentation



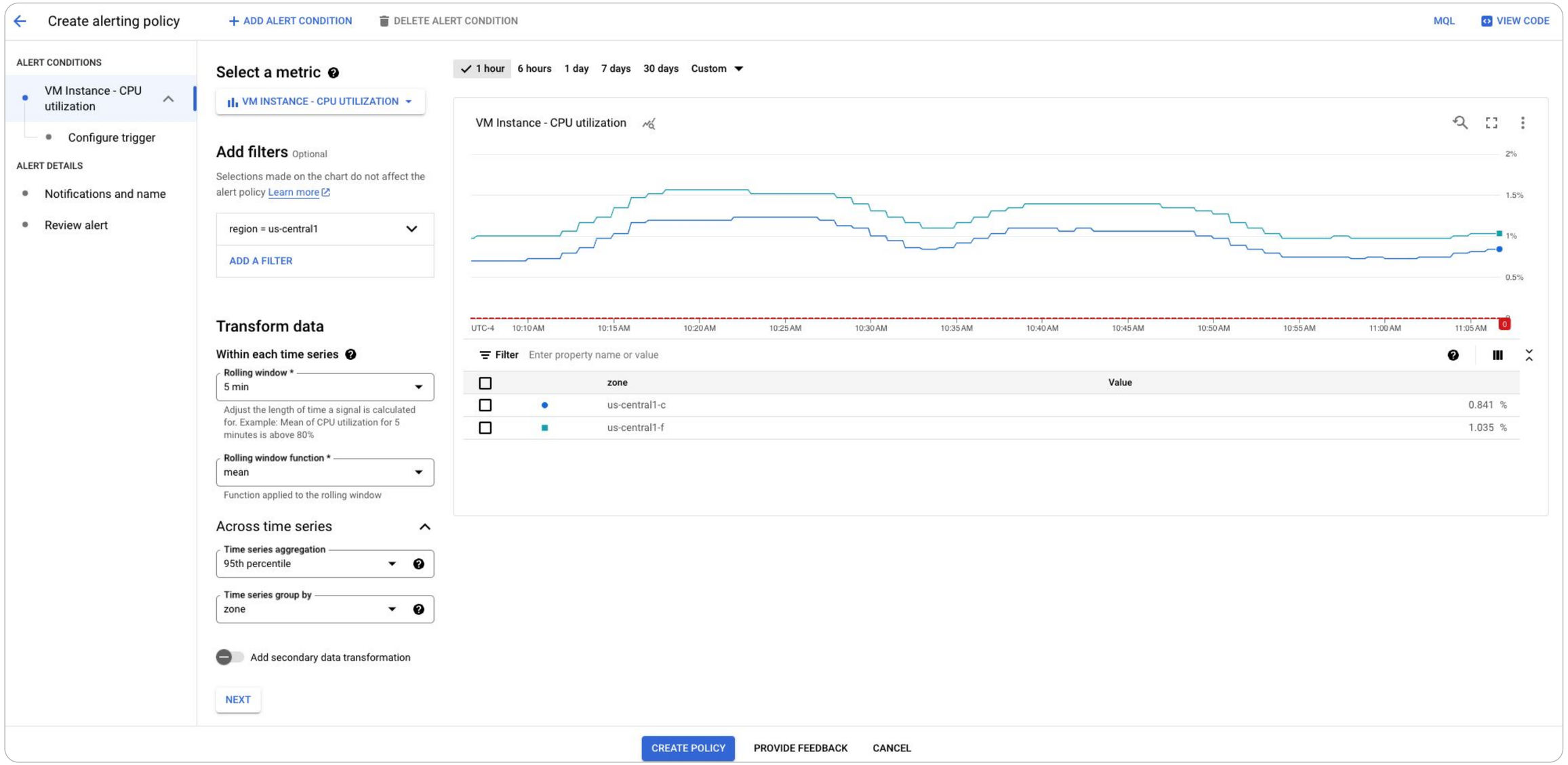
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Alerting policies are of two types

Metric-based alerting policies	Log-based alerting policies
Used to **track metric data collected** by Cloud Monitoring	Used to notify anytime a **specific message occurs in a log**,
Add a metric-based alerting policy by starting from the Alerts page of Cloud Monitoring.	Add a log-based alerting policy by using the Logs Explorer in Cloud Logging or under Cloud Monitoring
Example: Notify when the a**pplication that runs on a VM has high latency** for a significant time period.	Example: Notify when **a human user accesses the security key** of a service account.

Alert conditions

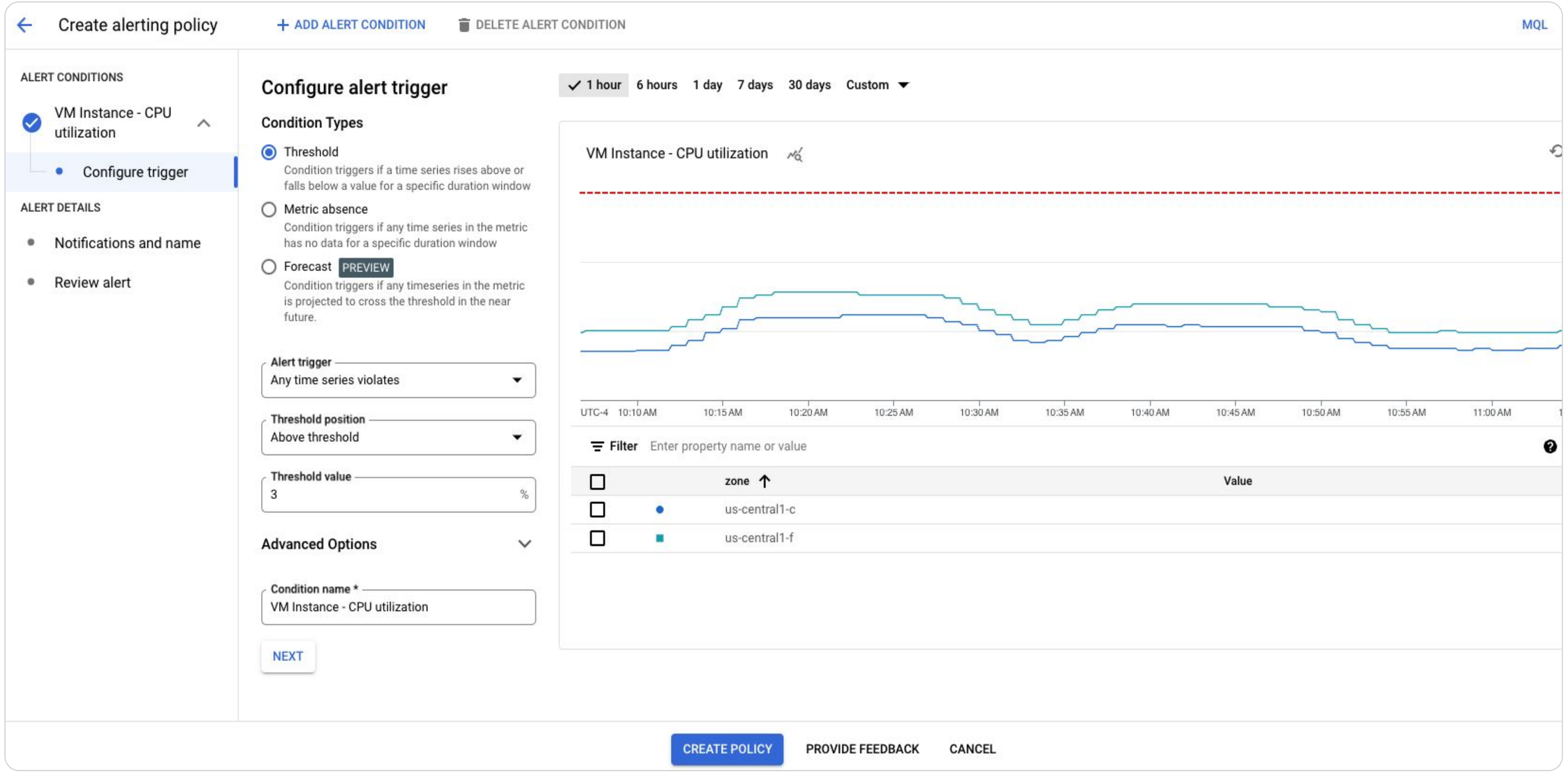
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Google Cloud

Configure alert triggers

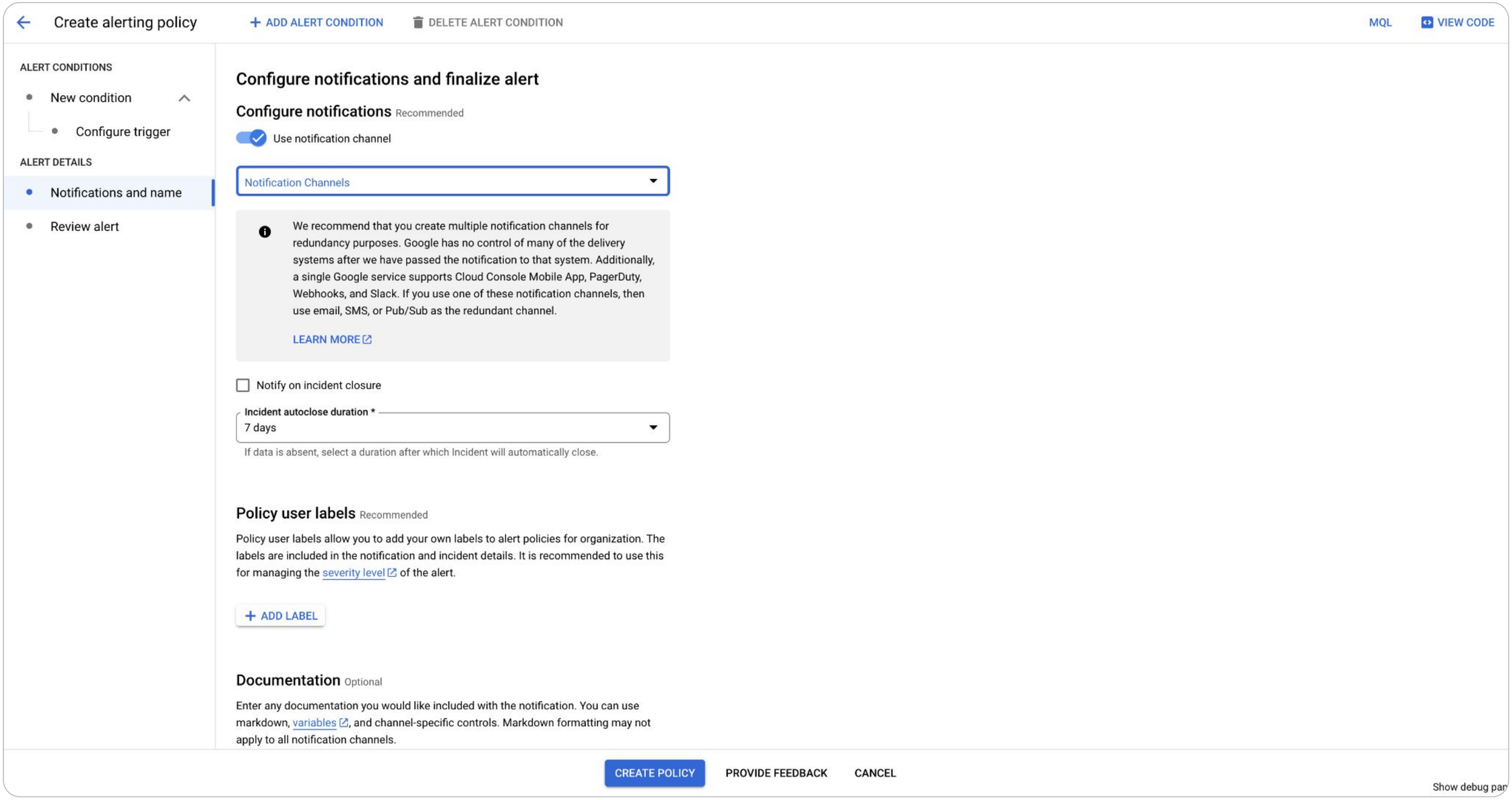
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Google Cloud

Configure notifications

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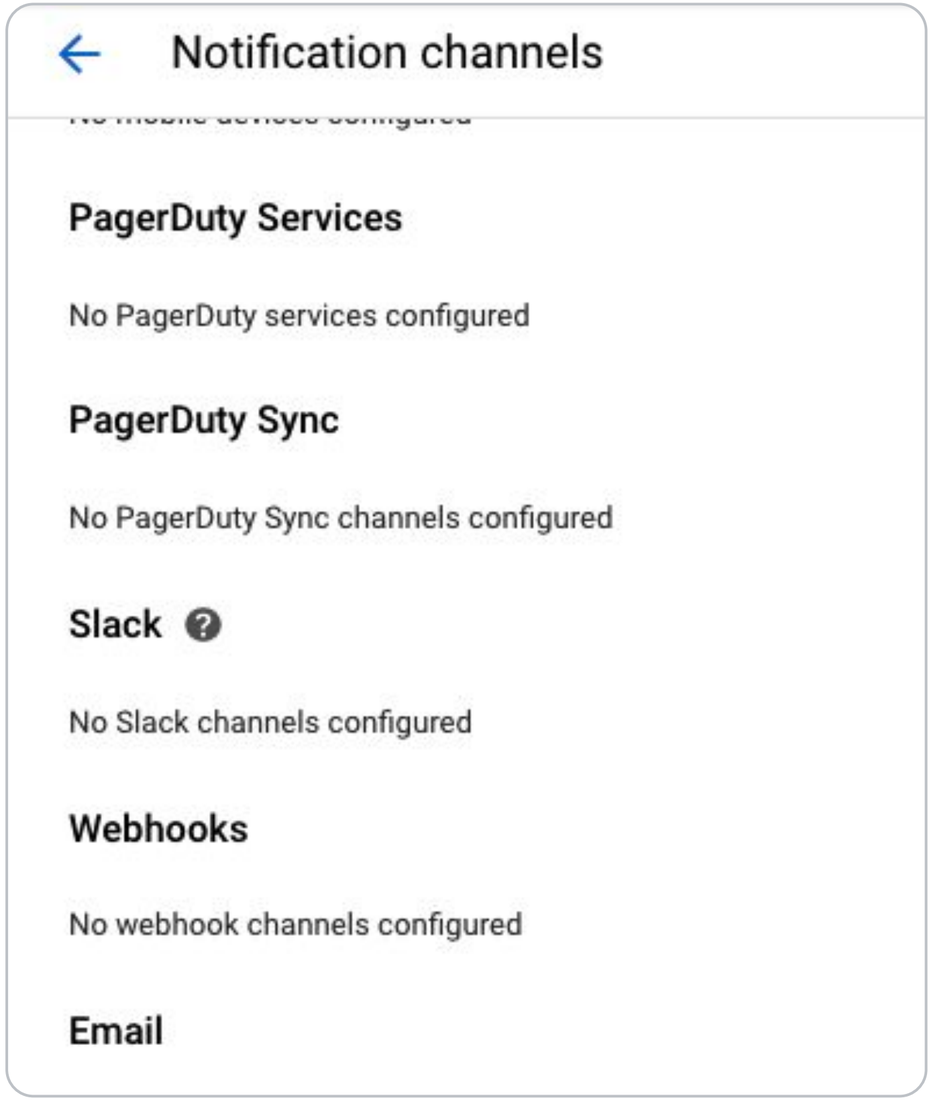
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Select notification channels

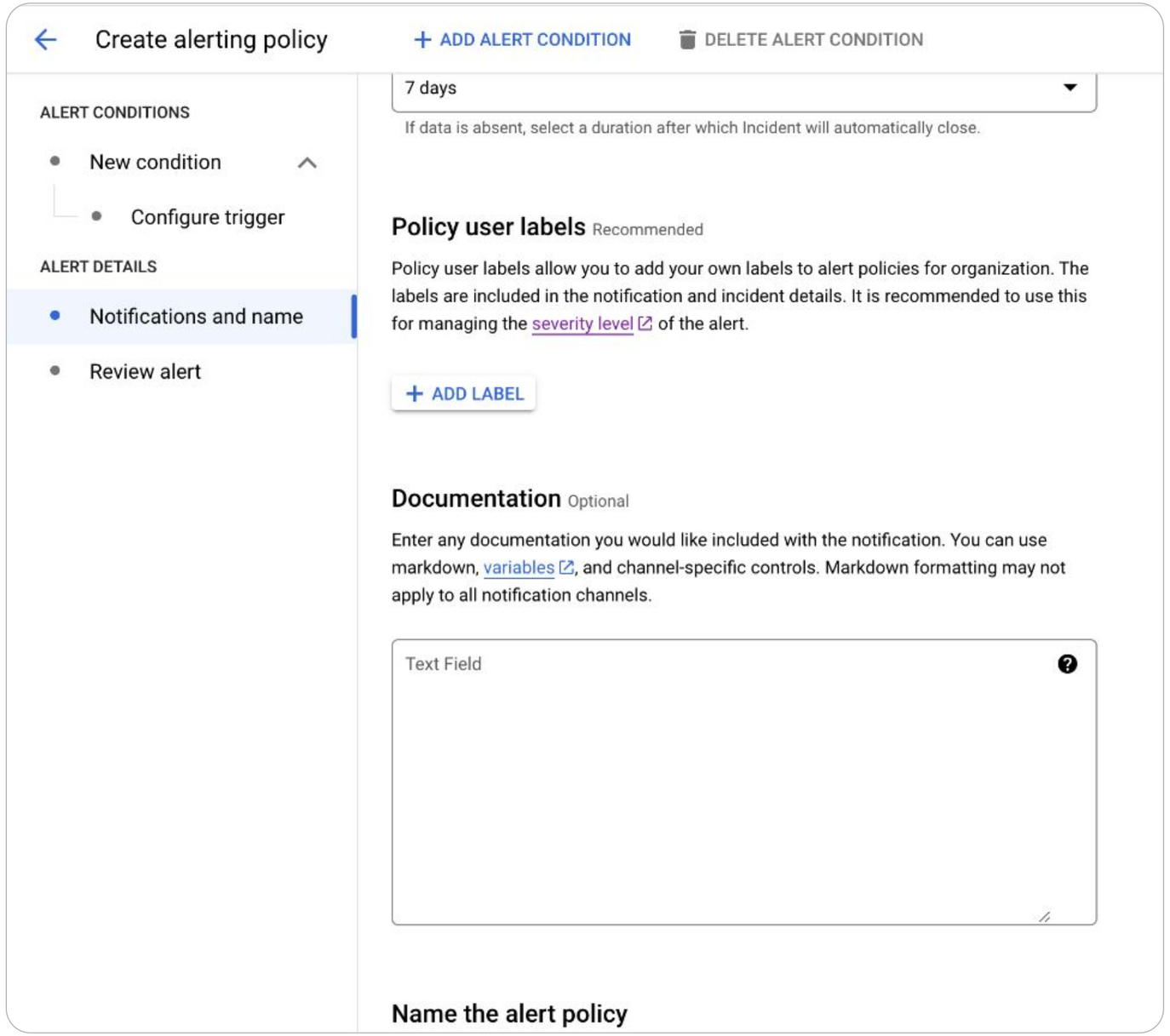
Supported notification channels include:

- Email
- SMS
- Slack
- Google Cloud app
- PagerDuty
- Webhooks
- Pub/Sub



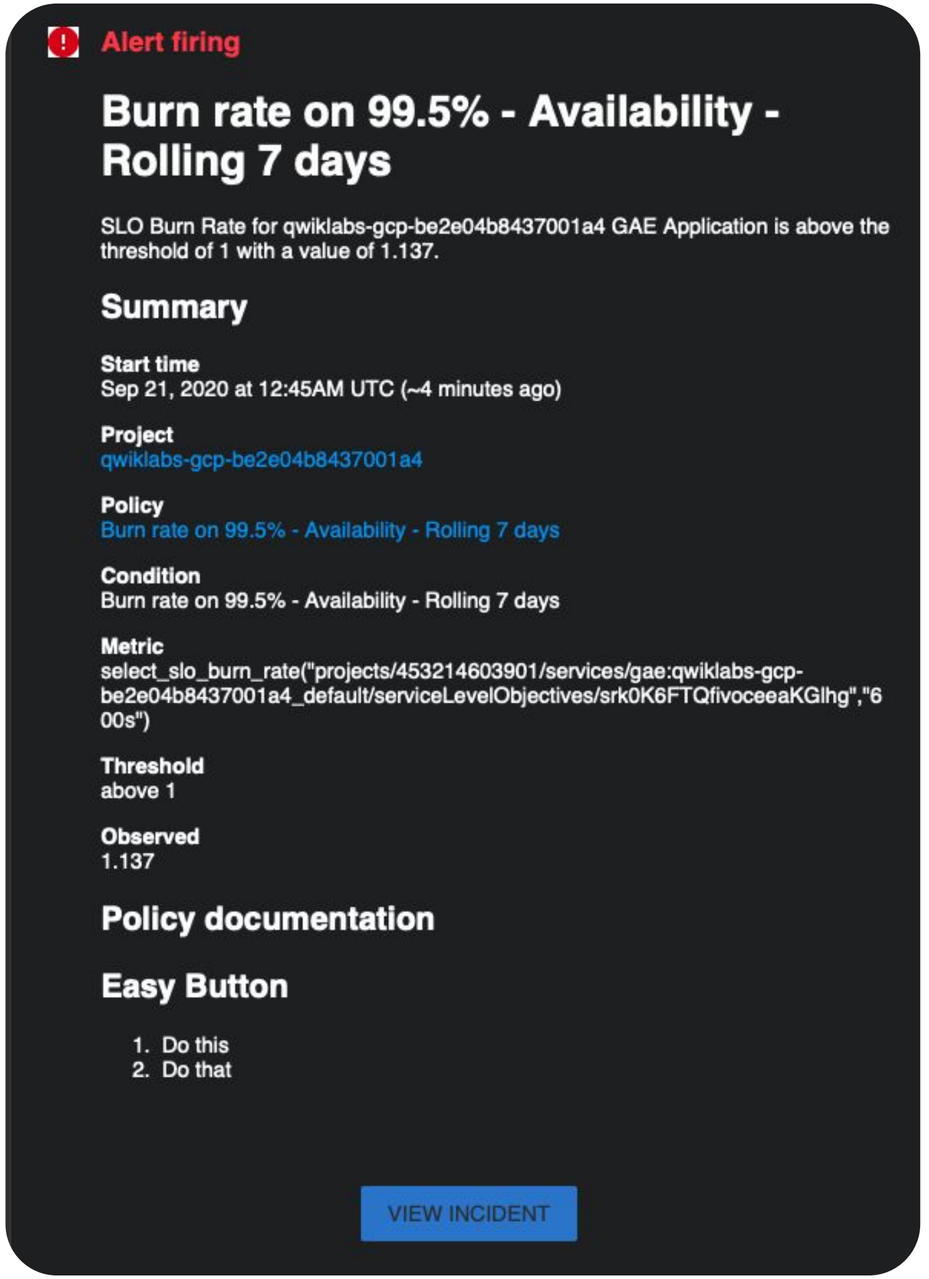
Documentation to guide troubleshooting

- Use the documentation section to guide your troubleshooting.
- Include internal playbooks, landing links and dynamic labels.
- Make it easy for the team to understand what is wrong.



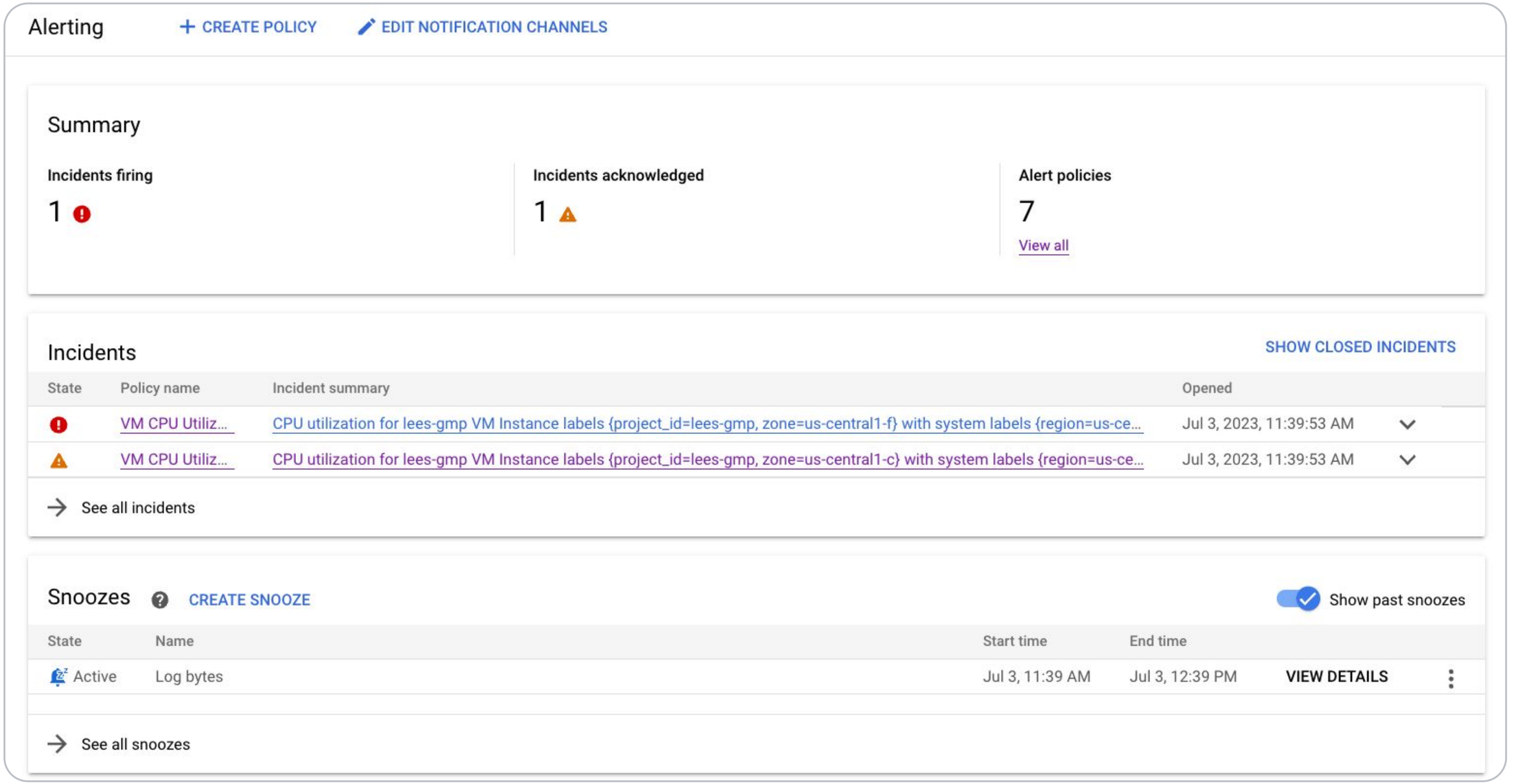
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Email alert example



Alerting web interface

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Resource groups can monitor multiple resources

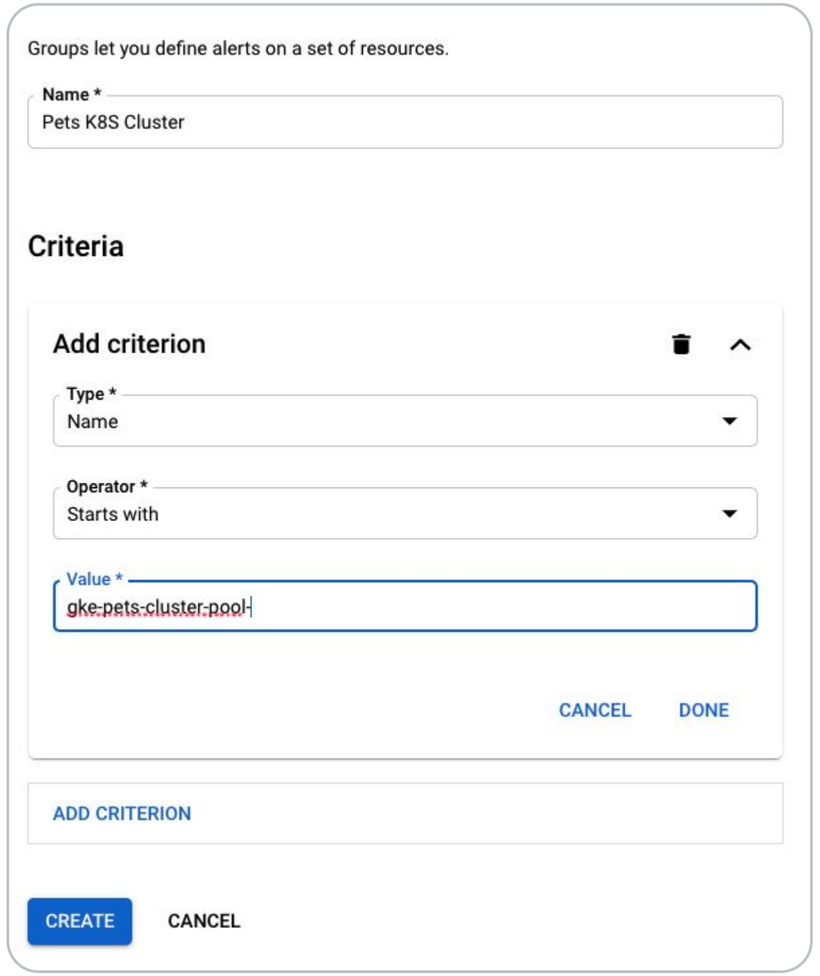
- Trigger based on the group instead of on individual resources.
- Groups can contain subgroups up to six levels deep.
- Resources can be members of more than one group.

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Use multiple criteria to create resource groups

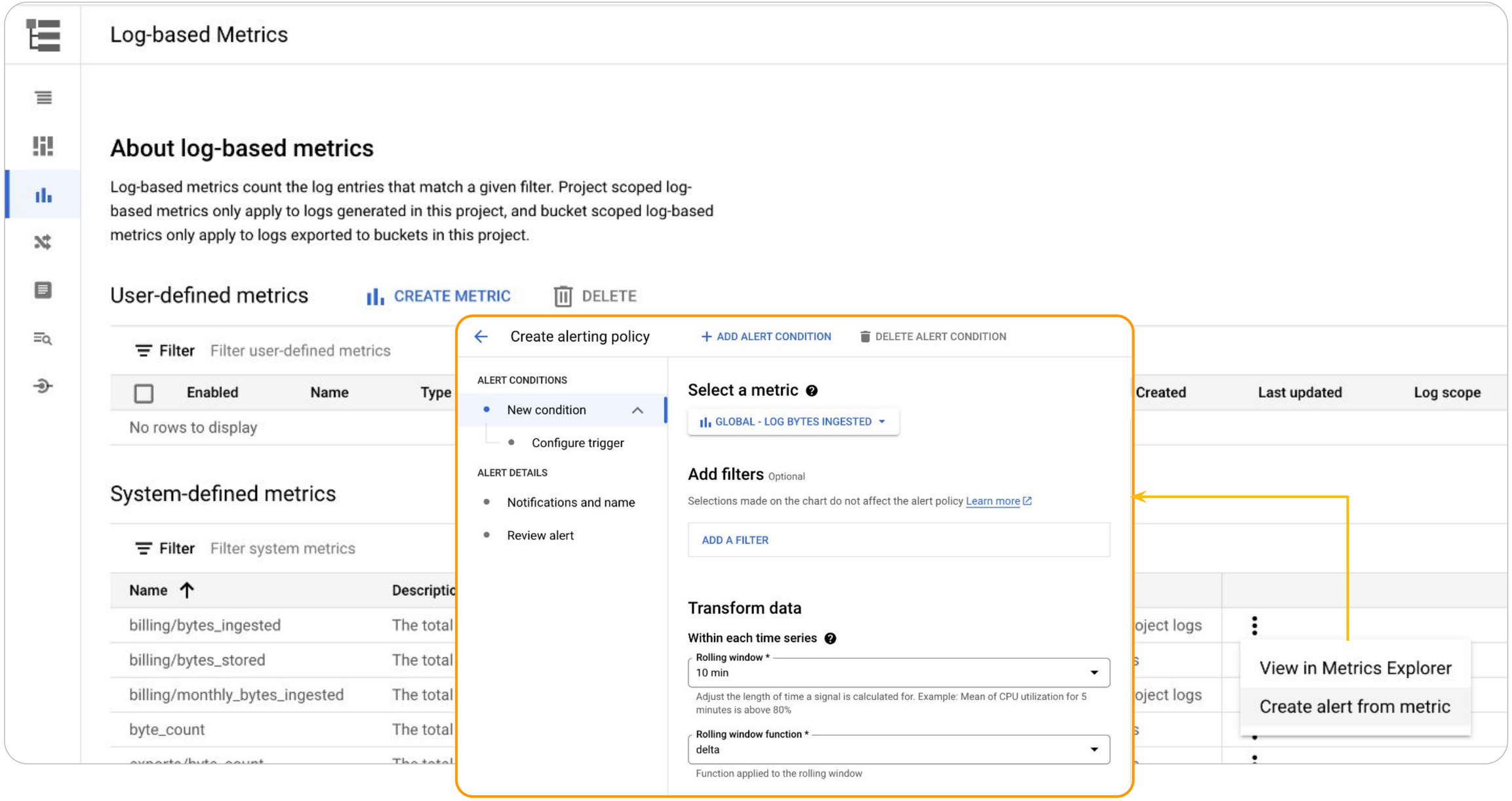
Criteria can include:

- Cloud Projects
- Resource name
- Resource type
- Tags and labels
- Security groups
- Regions
- App Engine apps and services



Attach alerts to logs-based metrics

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Create policies by using Terraform

```
resource "google_monitoring_alert_policy" "alert_policy" {
   display_name = "My Alert Policy"

   combiner     = "OR"

 conditions {
      display_name = "test condition"

      condition_threshold {

      filter     = "metric.type=\"compute.googleapis.com/instance/disk/write_bytes_count\" AND
      resource.type=\"gce_instance\""

      duration   = "60s"
      comparison = "COMPARISON_GT"
      aggregations {
      alignment_period   = "60s"
      per_series_aligner = "ALIGN_RATE"
      }
    }
  }
```

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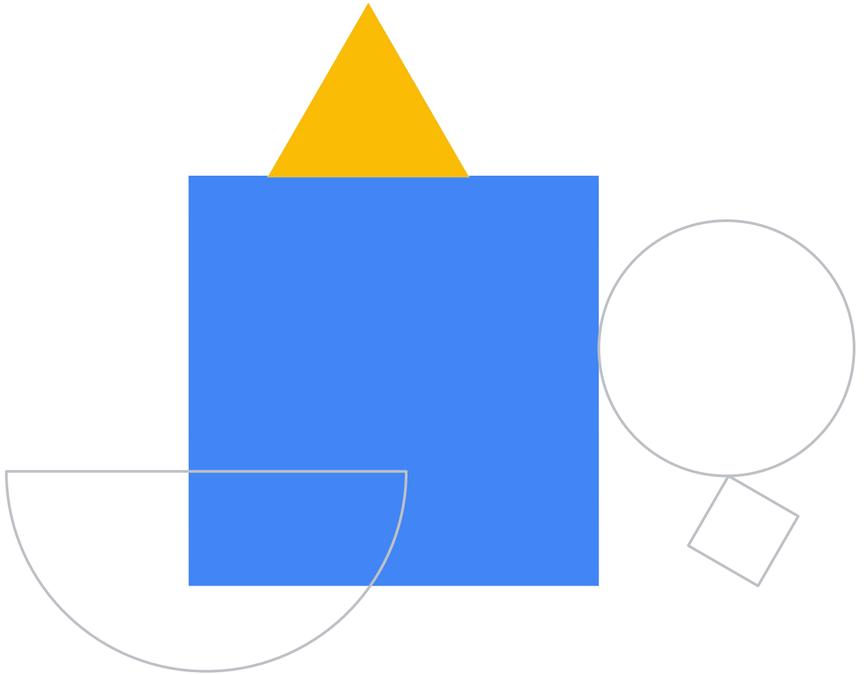
See documentation for more policy examples

- Group Aggregate Policy
- Uptime Check Policy
- Process Health Policy
- Metric Ratio Policy
- Setting for Common Alerting Policies
- Alerting JSON files

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Lab Intro

Alerting in Google Cloud



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In this section, you explore



- SLI, SLO, and SLA
- Developing an alerting strategy
- Creating alerts
- **Service Monitoring**

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Service Monitoring helps with SLO and alert creation

What are the **commitments regarding the availability and performance** of those services, and are your services meeting them?

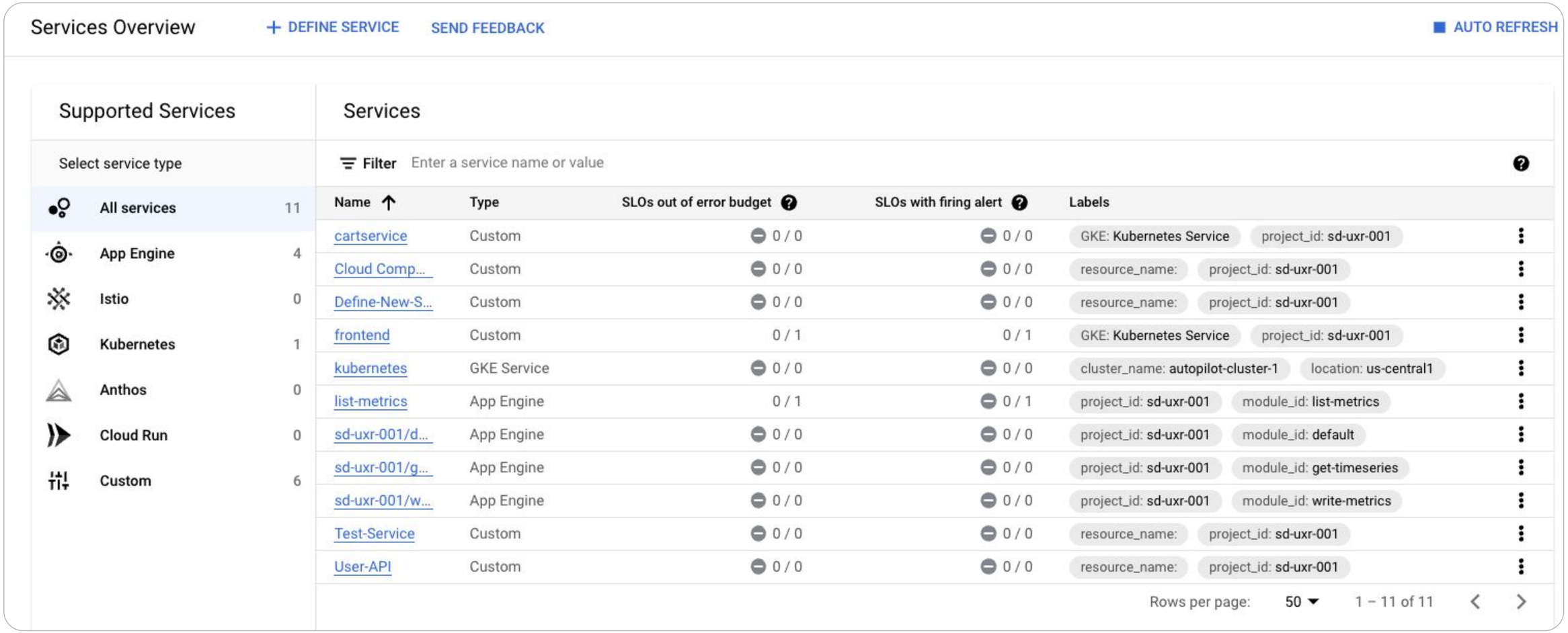
For microservices-based apps, what are the **inter-service dependencies**?

How to double **check new code rollouts and triage problems** if a service degradation occurs

Can you **look at all the monitoring signals for a service holistically** to reduce MTTR?

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Consolidated services overview



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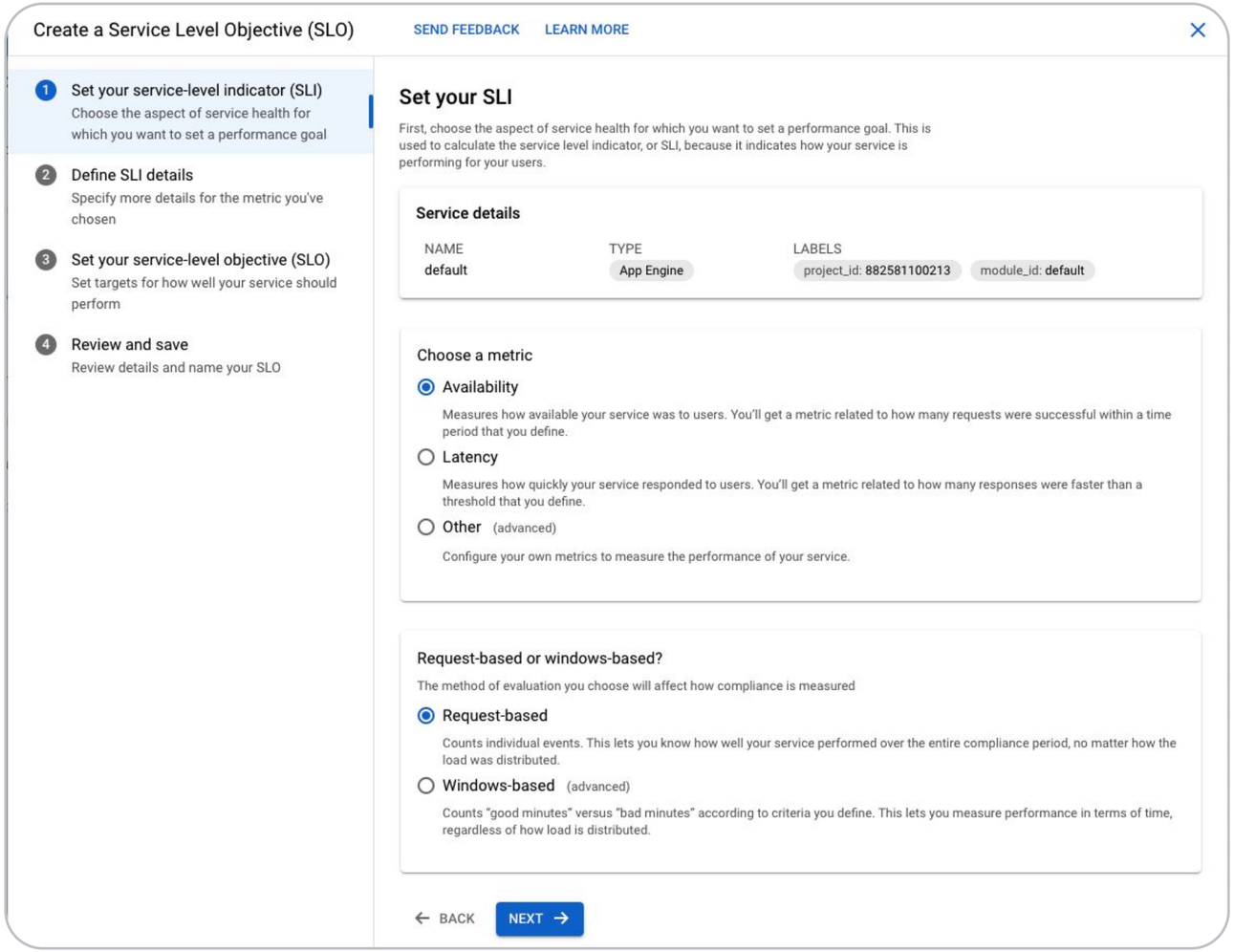
Windows-based versus request-based SLOs

- Imagine you get 1,000,000 requests a month and your compliance period is a rolling 30 days.
- A 99.9% request-based SLO can allow 1,000 bad requests every 30 days.
- A 99.9% windows-based SLO based on a 1-minute window can allow a total of 43 bad windows.
  - 43,200 total windows * 99.9% = 43,157 good windows.
- Windows-based SLOs can be tricky, because they can hide burst-related failures.

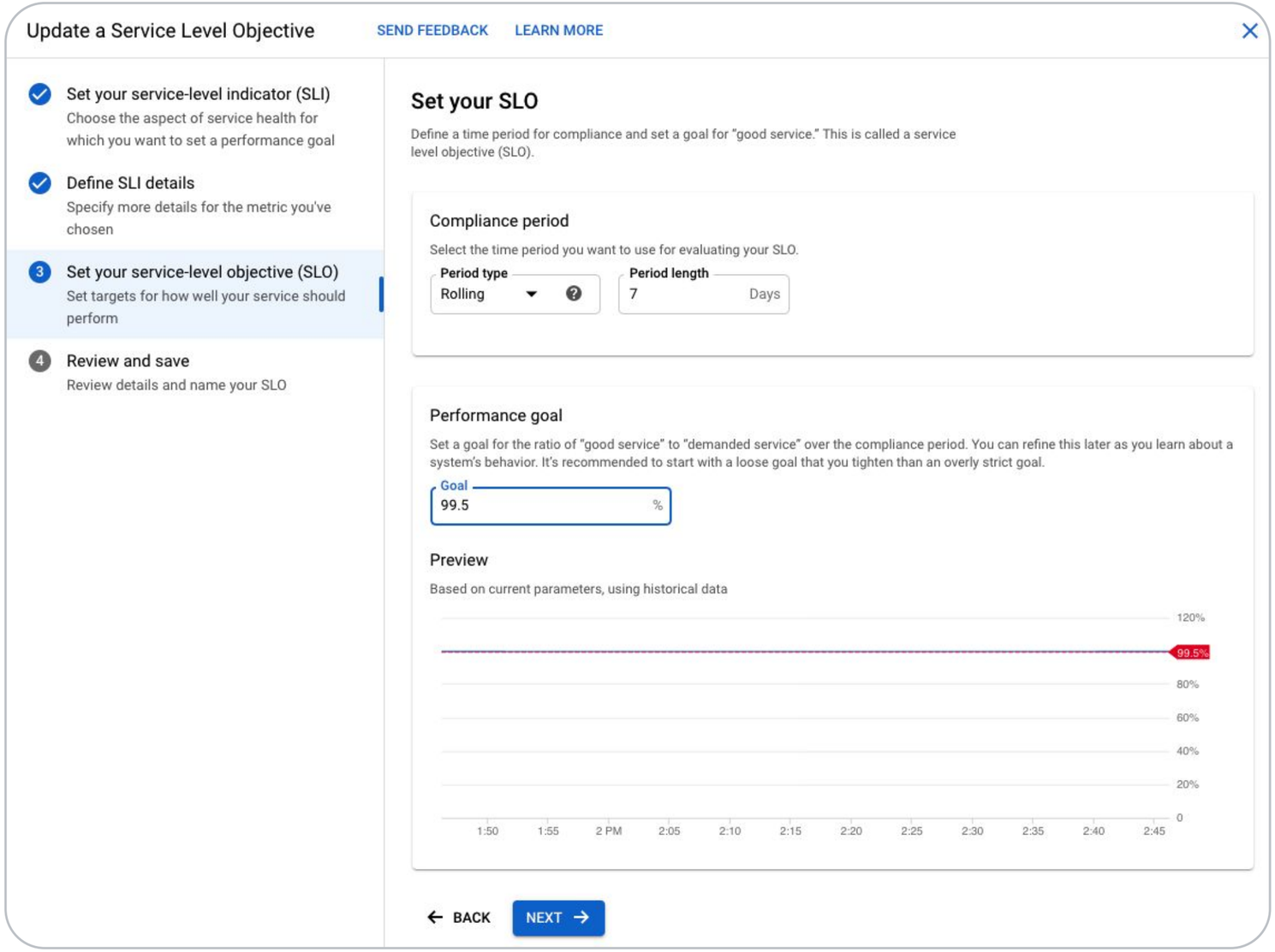
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Service Monitoring makes SLO creation easy

+ CREATE SLO

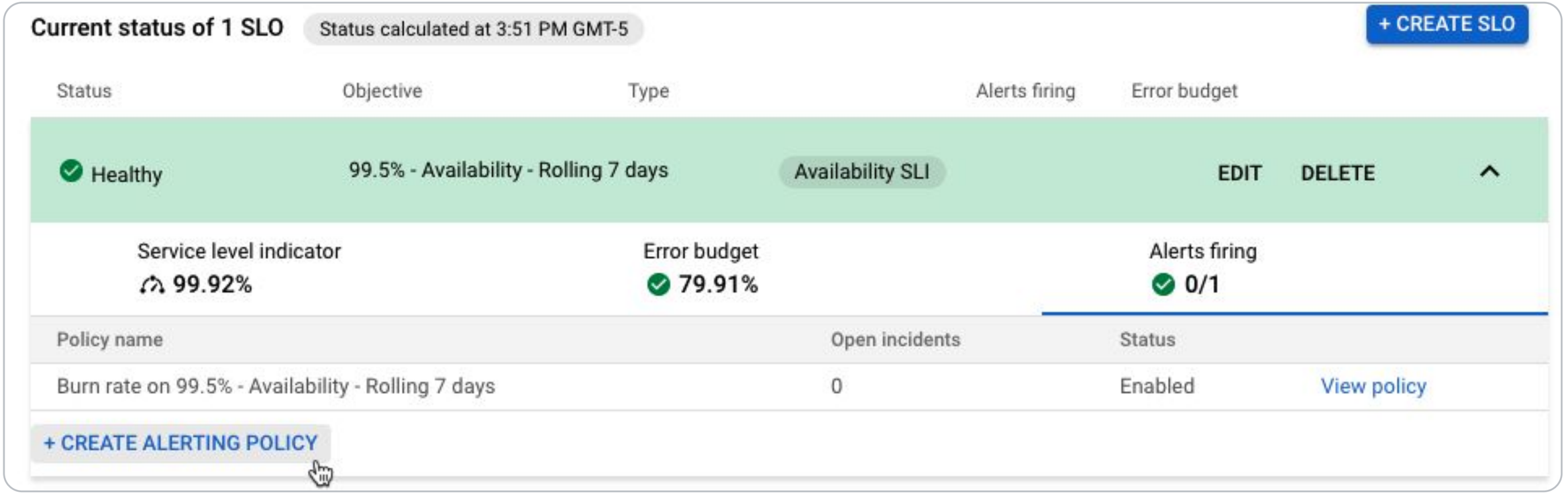


Set compliance period and goal



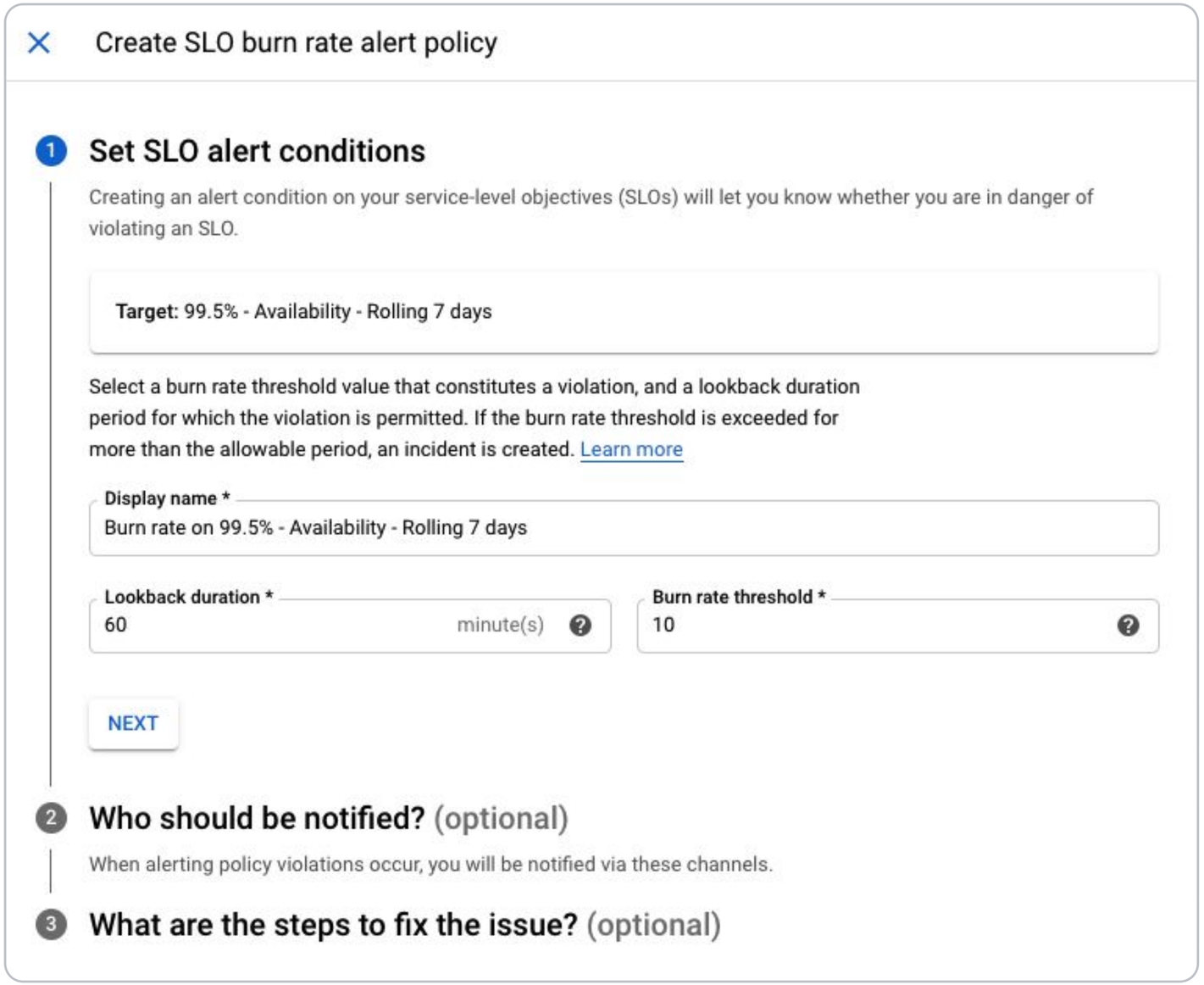
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SLO linked alerts are easy to create



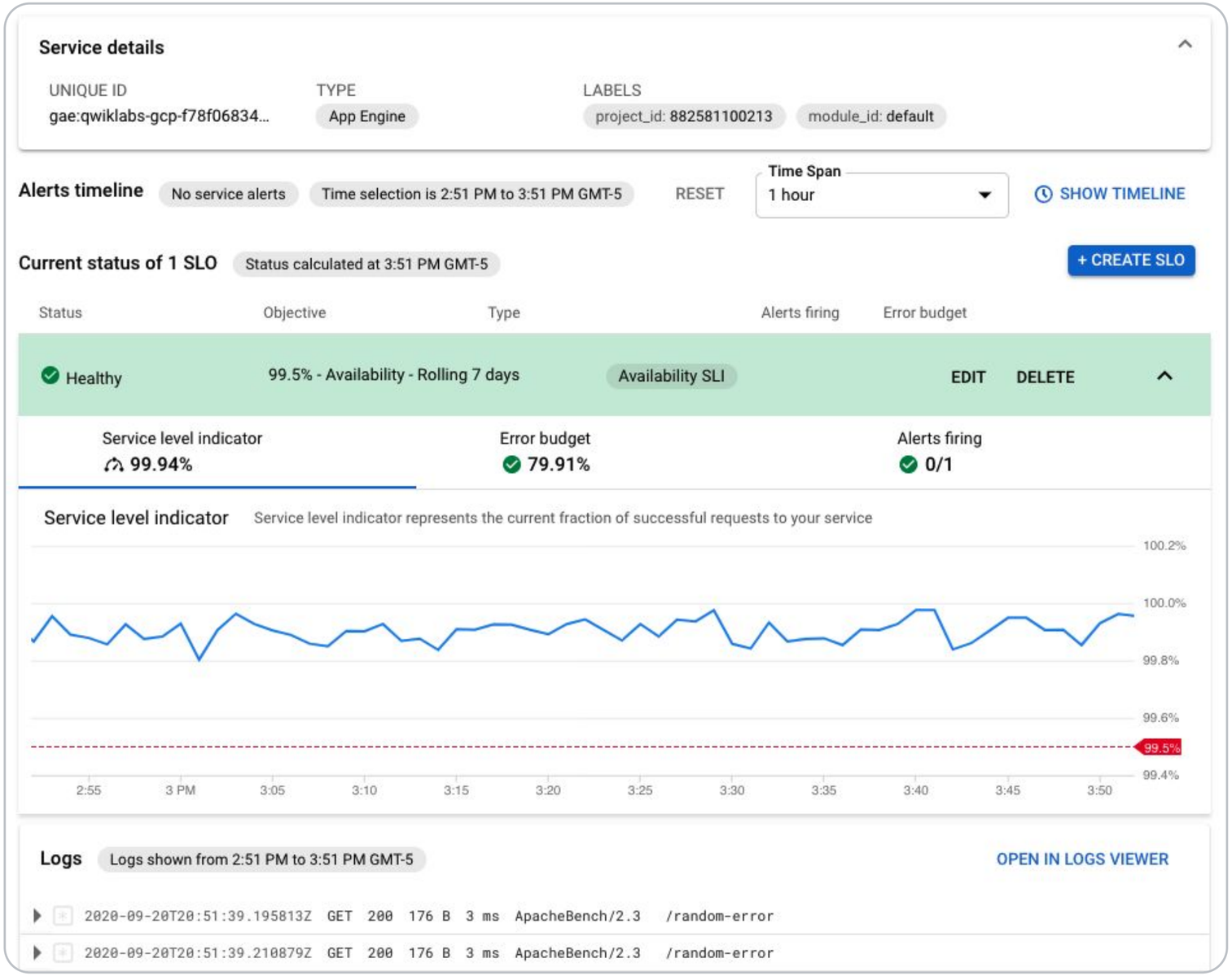
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Alert if the burn rate is too high



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SLO status is easy to monitor

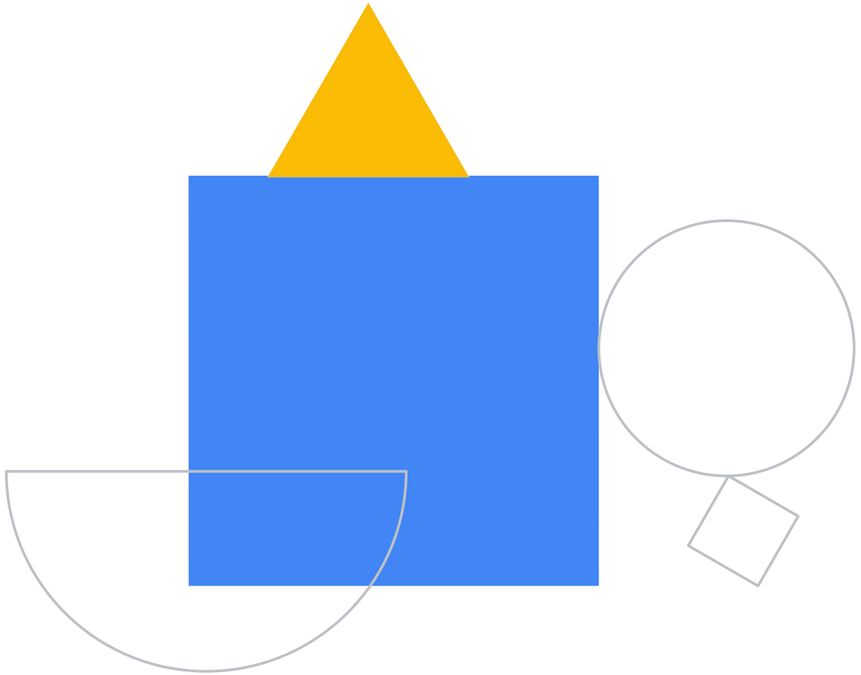


Google Cloud

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Demo

Using Service Monitoring



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Knowledge Check



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Quiz | Question 1

Question

In evaluating your alerting policies, which option best describes precision?

A. The proportion of events detected that were relevant to the sum of relevant and irrelevant alerts.

B. How long it takes to send notifications in various conditions.

C. How long alerts fire after an issue is resolved.

D. How long alerts take to be acknowledged.

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Quiz | Question 1

Answer

In evaluating your alerting policies, which option best describes precision?

A. The proportion of events detected that were relevant to the sum of relevant and irrelevant alerts. 

B. How long it takes to send notifications in various conditions.

C. How long alerts fire after an issue is resolved.

D. How long alerts take to be acknowledged.

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Quiz | Question 2

Question

Explain recall.

A. The proportion of alerts detected that were relevant to the sum of relevant and irrelevant alerts.

B. The proportion of alerts detected that were relevant to the sum of relevant alerts and missed alerts.

C. How long it takes to send notifications in various conditions.

D. How long alerts fire after an issue is resolved.

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Quiz | Question 2

Answer

Explain recall.

A. The proportion of alerts detected that were relevant to the sum of relevant and irrelevant alerts.

B. The proportion of alerts detected that were relevant to the sum of relevant alerts and missed alerts.



C. How long it takes to send notifications in various conditions.

D. How long alerts fire after an issue is resolved.

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Quiz | Question 3

Question

In the statement “Maintain an error rate of less than 0.3% for the billing system”, what is an SLI?

A. 0.3%

B. Error rate

C. Billing system

D. Less than 0.3%

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Quiz | Question 3

Answer

In the statement “Maintain an error rate of less than 0.3% for the billing system”, what is an SLI?

A. 0.3%

B. Error rate 

C. Billing system

D. Less than 0.3%

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Recap

01 Explain why SLI, SLO, and SLA are important.

02 Explain alerting policies and alerting strategies.

03 Explain error budgets.

04 Identify types of alerts and common uses for each.

05 Use Cloud Monitoring to manage services.



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